VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"Jnana Sangama", Belagavi- 590018

A Project Report
on

"Deep Learning and NLP-based Real-Time Sign Language Recognition and Contextual Conversion"

Submitted in partial fulfilment of the requirements for the award of the Degree of

Bachelor of Engineering

in

Computer Science and Engineering

Submitted by

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CERTIFICATE

This is to certify that the Project Report entitled **"Deep Learning and NLP-based Real-Time Sign Language Recognition and Contextual Conversion"** is a Bonafide Project work carried out by **KRUTHIKA D [1DB22CS072], MOHAMMED AJMAL HUSSAIN [1DB22CS089], PRAGATH R [1DB22CS109] and PRAJWALANISH V [1DB22CS110],** students of **Don Bosco Institute of Technology**, Bengaluru, in the partial fulfillment of 7th Semester for the **Degree of Bachelor of Engineering in Computer Science and Engineering** of **Visvesvaraya Technological University**, Belagavi, during the academic year 2025–2026. The Project Work has been appropriate as it satisfies the academic requirements in respect of the Project Work prescribed for the Bachelor of Engineering Degree.

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DECLARATION

We, **KRUTHIKA D [1DB22CS072], MOHAMMED AJMAL HUSSAIN [1DB22CS089], PRAGATH R [1DB22CS109] and PRAJWALANISH V [1DB22CS110]** students of seventh semester B.E, Department of Computer Science and Engineering, Don Bosco Institute of Technology, Bengaluru, declare that the Project Work entitled **"Deep Learning and NLP-based Real-Time Sign Language Recognition and Contextual Conversion"** has been carried out and submitted in partial fulfillment of the course requirements for the award of degree of **Bachelor of Engineering** in **Computer Science and Engineering** of **Visvesvaraya Technological University, Belagavi** during the academic year **2025-2026**. The Project Work has been approved as it satisfies the academic requirements for the award of Bachelor of Engineering Degree.

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ABSTRACT

Sign language serves as a vital communication medium for the Deaf and Hard-of-Hearing (DHH) community; however, the lack of technological support for regional sign languages such as Kannada Sign Language (KSL) creates significant communication barriers. Most existing sign language recognition systems focus on widely used languages and fail to address the linguistic complexity and contextual requirements of regional scripts. This project presents a deep learning and natural language processing (NLP)–based real-time system for recognizing Kannada Sign Language gestures and converting them into meaningful Kannada text and natural speech. The proposed system utilizes a Convolutional Neural Network (CNN) to recognize Kannada Vyanjanagalu (consonant) hand gestures from live webcam input. The recognized gestures are processed using a rule-based NLP module that performs Kannada-specific text construction, including vowel attachment, vattakshara formation, duplicate suppression, and automatic spacing to generate contextually correct text. A hybrid text-to-speech mechanism using the Web Speech API with Google Text-to-Speech (gTTS) fallback is integrated to produce clear and natural Kannada audio output. The system is implemented as a web-based application using FastAPI and WebSocket communication to enable low-latency real-time processing. Experimental results demonstrate high recognition accuracy and stable real-time performance under varying lighting and background conditions. The proposed approach highlights the effectiveness of combining deep learning and NLP techniques for regional sign language recognition and provides a scalable, cost-effective solution to enhance accessibility and inclusive communication for the DHH community.

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